

Resit Part 1 - Lectures 1 to 3

Instructions:

- The assignments are individual. You may consult and discuss with your colleagues, but you need to provide independent answers.
- You may use Matlab/Python/Julia to solve the assignment. Many questions require you clearly to do so.
- Provide detailed answers, describing the steps you followed.
- Provide clear reports, typed, preferably using LaTeX.
- Submit the reports digitally on Brightspace as indicated on the lectures.
- Include your code, for reproducibility of your results, in your Brightspace submission on a zip file, or through a link on your report. Upload your report on Brightspace in a separate PDF file.
- The code will *only* be used to verify reproducibility in case of doubts. The grading will be performed based on the results described in the report.

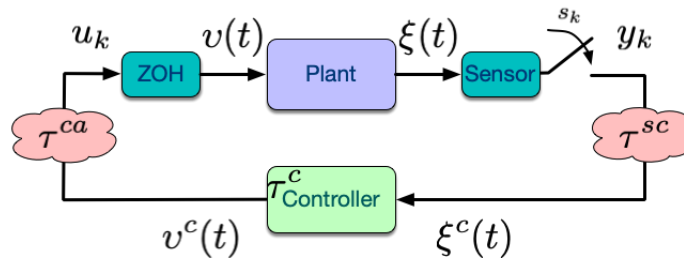


Figure 1: Schematic of a NCS including delays.

Consider a sample-data networked control system (NCS), as depicted in Figure 1. Assume that the plant dynamics are linear time-invariant given by:

$$\dot{\xi}(t) = A\xi(t) + Bv(t),$$

where the control signal v is a piece-wise constant signal resulting from the application of a controller in sampled-and-hold fashion, i.e. $v(t) = u_k, t \in [s_k, s_{k+1})$.

The system matrices are given by:

$$A = \begin{bmatrix} 0.5 - c & 0 \\ 0.2 + a - b & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where a is given by the first digit, b by the third digit, and c by the last digit of your student ID number.

We start considering the system with a constant sampling interval $h = s_{k+1} - s_k$ for all k , and assuming no delay is present $\tau = \tau^{sc} + \tau^c + \tau^{ca} = 0$.

Important note: For all the questions that follow, you do **NOT** necessarily need to provide analytic expressions, unless explicitly stated. However, you need to describe the steps you took to produce the resulting plots, and design process you followed.

Question 1: (1p)

1. (1p) Design a linear continuous time controller $v(t) = -\bar{K}\xi(t)$ placing the poles of the continuous closed-loop at $-2 + j$ and $-2 - j$. Construct the exact discrete-time model $x_{k+1} = F(h)x_k + G(h)u_k$, and the closed-loop system resulting from applying the controller in sample-data (piece-wise constant) fashion: $u_k = -\bar{K}x_k$, $x_k := \xi(s_k)$. Provide ranges of sampling times for which the sampled-data system is stable. Comment on what you observe.

Next we consider the case when delays are present in the networked system. Assume that the system is affected by a **constant** small delay $\tau \in [0, h)$, and it is controlled with the *static* controller $\bar{K}$.

Question 2: (4p)

1. (2p) Construct the exact discrete-time model for the NCS with delays and study the combinations of sampling intervals and system delays that result in an asymptotically stable closed-loop. Provide plots illustrating the combinations of (h, τ) retaining stability.
2. (2p) Select a sampling interval h that guarantees stability under no delays. Redesign the controller to improve the robustness against delays, that is, try to design a controller that increases the range of tolerable delays for the selected h . Think about designing now a simple *dynamic* controller instead.

Next we consider a different approach to deal with delays $\tau \in [0, h)$. If the delay experienced by an input upon arrival at the ZOH is $\tau \leq h/2$ the input is applied as usual, while if the input arrives with a delay $\tau \geq h/2$ the input is ignored and the previous input is held.

Question 3: (7p)

1. (2p) Model mathematically the described situation.
2. (1p) Without any additional assumptions on the delay sequences, can you argue (without solving any LMIs) whether the resulting system can be stable?
3. (1p) Verify your previous answer employing LMIs.
4. (2p) Let the sequence of delays be given by $(\tau_1 \tau_2 \tau_3)^\omega$, with $\tau_1 = h/3$, $\tau_2 = 3h/4$, $\tau_3 = h/4$. Derive sufficient conditions to establish the stability of the resulting closed-loop system.
5. (1p) Provide a plot depicting the range of values for the sampling time which result in stability of the closed-loop.